

Alexander Engelhardt

Earth - Technology - Arts

For the past decade I have actively built the skills and experience to function in a cross-domain role combining technology, communication, strategy and the world. This originates in a deep appreciation of our society and I want to be able to contribute and protect it while building groundbreaking technology.

I chose to study computer science and AI because understanding technology is a fundamental part of understanding today's world. For the past two years I have worked in a role that drew on all my skills, from research and engineering to managing a team, working with executives to understand their organisation's AI needs, and designing the system architectures to meet them.

EXPERIENCE

AgentFormers – Research to Business, 2025 - Present
Business Finland funded, University of Helsinki

Leading product, commercial and day-to-day project management for AgentFormers, an R2B project building a sovereign AI-agent ecosystem that runs inside an organisation's own infrastructure, under its own policies, with a full record of what every agent does. While my original role in the project was focused more on research and engineering, I saw an opportunity to strengthen coordination and commercial activity in the group, and took it on.

- **Product and solution design:** Owning the roadmap from governed agent runtime to a sovereign AI workspace and domain applications, with model routing across Azure, CSC (Finland's national supercomputing centre) and on-premises hardware.
- **Hands-on development:** Co-architecting and implementing the AgentFormers platform and application layers, translating customer requirements directly into implementation while running the product and commercial side.
- **Executive discovery:** Leading customer interviews and workshops with organisations across universities, cities, finance, industry and life sciences, in Swedish, Finnish and English. Turning the findings into product priorities and go-to-market strategy.
- **Business cases and ROI:** Build market and revenue models, pricing logic and procurement-aware pilot structures for public and regulated buyers. Map EU AI Act, GDPR, DORA/NIS2 and ISO 27001 requirements into the offer.
- **In-house adoption:** Coordinating together with UH Head of IT, CEO of Helsinki Innovation Services and Director of Innovations and Ecosystems to draw up an architecture that allows for easy agent adoption while accommodating 10+ years data residency.
- **Some of the applications in production:**
  - **Missions Cowork:** A sovereign alternative to Microsoft Copilot/Claude Cowork/Mistral Vibe with around 30 test users across five domains in its first two weeks, including a dedicated tenant for ILaaS, the French higher-education inference federation. Uses models from different providers (Azure, sovereign cloud) or self-hosted including Mistral and Leanstral.
  - **Innovation application:** An application for mapping out opportunities for funding, commercialisation and consortium building across academia and industry.
  - **Advisory agents:** piloted with a municipal business services in Helsinki and Tampere.
  - **Secure proof-assistant agents for mathematics research:** An agent and orchestration framework for off-loading sub-agents to CSC compute and Lean compilers. In use by mathematics professors at four universities today.
- **Partnerships and pitching:** Pitch to numerous investors and present at conferences and events. Wrote the investment memorandum.

Neural Publish-Subscribe for 6G, Master's Thesis, 2024-2025
Business Finland funded, University of Helsinki

Developed an intelligent LLM-based publish-subscribe system for matching data across domains. This includes the matching algorithms itself as well as a system for deploying the system components across a distributed network of nodes. Awarded the highest grade. The work is the basis for a patent pending.

Innotin - LLM-based Innovation Service, 2024
Toska Software Engineering Group

At the Software Engineering Group Toska, I built Innotin, an LLM-based service that helps researchers turn their research into invention disclosures. Beyond this, my role in the group became one that was taken to client meetings due to my excellent communication skills and ability to connect with people and bring out innovative ideas that weren't on the clients mind at first.

Project Coordinator: MammalBase, 2023
University of Helsinki & The Finnish Museum of Natural History

Software Engineer & Artist: Berry Picker Tracker Mobile App, 2023
University Of Helsinki

Teacher, 2018-2020
Granhult, Lagstad, Rödskog

It remains one of my most valuable management experiences — if you can engage and manage 25 children, you can manage anything.

EDUCATION

PhD in Computer Science (on going)
University of Helsinki (2025)

After my role in the Neural Publish-Subscribe project my supervisor invited me to do a PhD within AgentFormers project. I naturally accepted. As the project's commercial needs grew, I chose to focus on product and customer work, which meant deprioritising my PhD. I deeply love research and science, but this shift made me realise what role really speaks to me at this current moment.

Master's Degree in Computer Science
University of Helsinki (2024-2025)

I finished my Master's studies with a mean grade of 4,87/5 and the highest grade possible (5/5) in my Master's thesis, *Neural Router: A System for Structure-Agnostic and Configuration-Free Content Delivery*. This includes the design of a larger agentic pub-sub architecture and an LLM-based algorithm for matching data without fixed data-structures or need for configuration. The thesis was used as a ground for a patent that is currently pending.

Bachelor's Degree in Computer Science
University of Helsinki (2021-2023)

I finished by Bachelor's studies with a mean grade of 4.17/5 and the highest grade possible (5/5) in my Bachelor's thesis, *Comparing Monolithic- and Microservice Architecture through the lens of Quality Attributes*.

Minor in Geography and Geoinformatics
University of Helsinki (2020-2021)

I completed a minor in geography with over 60 credits. This included physical and human geography, urban planning, and geoinformatics. There is no discipline as interdisciplinary as geography where you get to understand how mountains turn to sand and how sand turns into high-rise buildings. These buildings, in turn, can lead to economic wonders or the collapse of ecosystems.

HOW I WORK

- I can both be led or take leadership depending on circumstances.
- No pride - low ego
- I make customer and partners feel that we are on the same team solving a shared problem.
- Solve the problem - not the task.
- I try to make people achieve their best potential

WHAT I VALUE

- Team work and collaboration
- Openness and honesty
- Courage to stand up for something
- Willingness to commit fully to things
- Culture where people do not fear expressing ideas that might seem "dumb"
- Humour in stressful environments

GENERAL SKILLS & INTERESTS

- Teamwork and Leadership
- Communication and Sales
- Product development
- Agentic vision and where are we in 1, 5, 10 years?
  - Different levels of agent autonomy
  - Interactive cowork vs. Function automations
  - Cognitive debt and agents
  - How to present agents to people?
  - Single-agent vs. Multi-agent?
  - Personification of agents
- Society, history, democracy & philosophy
- Writing
- Music
- Film
- Art

TECHNICAL SKILLS

- Agents:
  - How LLMs work as part of agents
  - Ollama, vLLM
  - Context management
  - Harness Engineering
  - Meta-harness engineering
  - Policy Control
  - Tamper-evident auditing
  - Model Routing
  - Distributed deployment patterns
- Full-Stack engineering
- Publish-subscribe systems
- Microservice architectures
- Distributed systems
- Unix.

LANGUAGES

Swedish: Mother tongue, Native
English: Fluent
Finnish: Fluent

CONTACT

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